

GNIOT Institute of Management Studies – PGDM (Batch 2022-24)**(TRIMESTER –IV) END TERM EXAMINATION, OCTOBER-2023****DATABASE MANAGEMENT SYSTEM****[Time: 2 Hours 30 Minutes]****[Maximum Marks: 70]**

[**BL** – Bloom's Taxonomy Levels (1- Remembering, 2- Understanding, 3 – Applying, 4 – Analysing, 5 – Evaluating, 6 - Creating)
CO – Course Outcomes]

Following are the course outcomes of the subject: - Database management system

CO	Course Outcome (CO)
CO1	Be able to understand the importance of DBMS in real life application.
CO2	Understanding the ER diagram business model and its application.
CO3	Apply query processing techniques to automate the real time problems of database.

(Note- Faculty can add the more CO's as per their subject.)**SECTION-A****Attempt all parts of the following: (Short Answer Type Questions, do not exceed 200 words)**

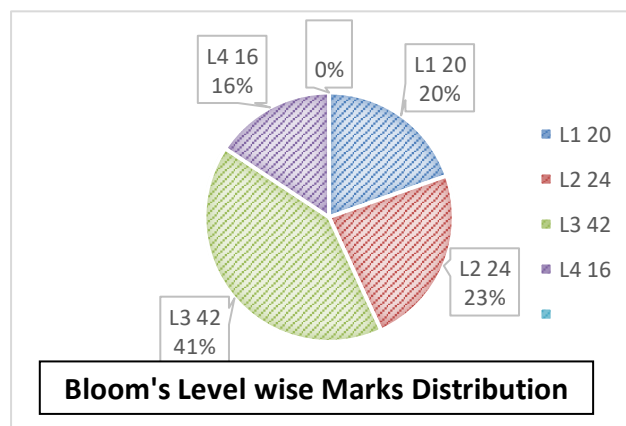
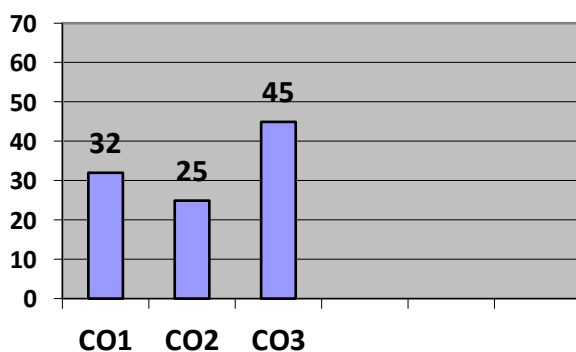
Q.1	Marks	CO	BL
A	4	CO - 1	L - 1
B	4	CO - 1	L - 2
C	4	CO - 3	L - 3
D	4	CO - 2	L - 3
E	4	CO - 2	L - 2

SECTION-B**Attempt any 3 questions out of the following: (Long Answer Type Questions)**

Q. No	Marks	CO	BL
2 (A)	8	CO - 1	L -2
2 (A)	8	CO - 1	L -1
2 (B)	8	CO - 1	L -4
2 (B)	8	CO - 3	L -3
3 (A)	8	CO - 3	L -4
3 (A)	8	CO - 3	L -3
3 (B)	8	CO - 3	L -2
3 (B)	8	CO -2	L -1

SECTION-C**Attempt all parts of the following: (Caselet / Numerical)**

Question	Marks	CO	BL
4 (A)	9	CO - 3	L - 3
4 (B)	9	CO - 2	L - 3

Course Outcome Wise Marks Distribution

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(TRIMESTER – IV) END TERM EXAMINATION, OCTOBER - 2023
DATABASE MANAGEMENT SYSTEM

[Time: 2 Hours 30 Minutes]

[Maximum Marks: 70]

INSTRUCTIONS:

- (i) There are **THREE** sections in this question paper.
- (ii) Attempt questions from each section as per the directions.
- (iii) Make suitable assumptions wherever necessary.

SECTION-A

(Short Answer Type Questions, do not exceed 200 words)

Q.1. Attempt all parts of the following:

(5×4=20)

- (a) Define database and database management system.
- (b) Discuss how candidate key is different from super key.
- (c) Differentiate between DDL and DML.
- (d) Construct an ER diagram to barrow the loan from the bank.
- (e) Define an entity. Mention the types of entity.

SECTION-B

(Long Answer Type Questions)

Q.2. Attempt all questions of the following

(4×8=32)

- (a) Explain the use of 'Foreign Key' in a Relational Database Management System. Give example to support your answer.

OR

- (a) List and explain the different types of database users.
- (b) State four advantages of DBMS over file processing system.

OR

- (b) Discuss selection and projection operation in relation algebra with an example.

Q.3. (a) Illustrate the following commands DDL and DML: INSERT, UPDATE, ALTER, DROP

OR

- (a) Define join operation. Illustrate different types of joins with example.
- (b) Discuss different types of relational algebra operations.

OR

- (b) Explain the E-R diagram notations used in database.

SECTION-C

Q.4. Attempt all parts of the following:

(2×9=18)

4(a) Write the outputs of the SQL queries (i) to (vii) based on the relations COMPUTER and SALES given below:

Table: COMPUTER

PROD_ID	PROD_NAME	PRICE	COMPANY	TYPE
P001	MOUSE	200	LOGITECH	INPUT
P002	LASER PRINTER	4000	CANON	OUTPUT
P003	KEYBOARD	500	LOGITECH	INPUT
P004	JOYSTICK	1000	IBALL	INPUT
P005	SPEAKER	1200	CREATIVE	OUTPUT
P006	DESKJET PRINTER	4300	CANON	OUTPUT

Table: SALES

PROD_ID	QTY_SOLD	QUARTER
P002	4	1
P003	2	2
P001	3	2
P004	2	1

1. SELECT MIN (PRICE) , MAX (PRICE) FROM COMPUTER;
2. SELECT COMPANY, COUNT FROM COMPUTER GROUP BY COMPANY HAVING COUNT (COMPANY) > 1;
3. SELECT * FROM COMPUTER WHERE PRICE > 800;
4. UPDATE SALES SET QTY_SOLD ='7' WHERE PROD_ID=P003;
5. DELETE FROM SALES WHERE PROD_ID='P004';
6. SELECT PROD_ID, PRICE FROM COMPUTER WHERE TYPE='INPUT';
7. SELECT PROD NAME, COMPANY, QUARTER FROM COMPUTER C, SALES S WHERE C. PROD ID=S. PROD ID;

4(b) Design an ER model of university database application considering the constraints –

- A university has many departments.
- Each department has multiple instructors (one person is HOD). Here the HOD refers to the head of department.
- An instructor belongs to only one department.
- Each department offers multiple courses, each subject is taught by a single instructor.
- A student may enroll for many courses offered by different departments.